



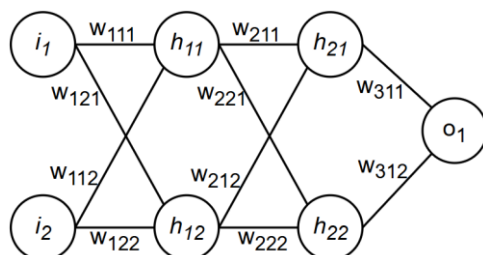
NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

Department	Computer Science & Engineering
Examination	Mid Semester
Program	B.Tech/PhD
Session	Autumn
Course code & Course Name	CS6218 (Machine Learning)
Marks	30
Duration	2 hours

There are ... 5 ... questions printed on ...02... pages

Instructions: 1. In case of any doubt, write your assumptions, write it clearly and continue.
2. Attempt all the questions.

Q. No.	Questions	Marks	CO*, BL**
1.	You are given a dataset of house prices with features (area, number of rooms, age of house, location rating) and a binary label (High-priced = 1, Low-priced = 0). (A) Explain the differences between classifiers implemented via (i) linear regression, (ii) logistic regression, focusing on objective functions and training algorithms. (B) Why is the AUC-ROC curve a more informative metric than accuracy for binary classification tasks? Explain why the F1-score is considered a better metric than accuracy in some classification tasks.?	3 3	CO1 L4,5
2.	Consider the neural network shown below: - Inputs: $i_1=0.1$, $i_2=0.2$ - Desired (actual) output: 0.90 - Hidden layer nodes with Relu activation - Output node (o_1) with sigmoid activation - Initial value for all weights =0.1 (A) Perform a forward pass through the network and compute the total error. (B) Derive the weight update equation and perform a single step of backpropagation to update the weights w_{111}, w_{311}, and w_{211}.	3 3	CO1,2 L5,6



3.	Compute the LDA to project the following data points of two classes: Class 1: $x_1=[2,3]$, $x_2=[3,3]$, $x_3=[2,2]$ Class 2: $x_4=[7,8]$, $x_5=[8,8]$, $x_6=[7,7]$	3	CO 2 L3,4																												
(A)																															
(B)	A student needs to make a decision on which courses to take, based only on the first lecture's impression. The prior probability and class-conditional probability are given in the following table: Apply Bayes' Theorem to make decisions on $P(\text{good} \text{interesting lecture})$ and $P(\text{fair} \text{interesting lecture})$ <table><tr><td>$p(x/w_i)$</td><td>good</td><td>fair</td><td>bad</td></tr><tr><td>Interesting lecture</td><td>0.8</td><td>0.5</td><td>0.1</td></tr><tr><td>Boring lecture</td><td>0.2</td><td>0.5</td><td>0.9</td></tr></table> <table><tr><td>Quality of the course</td><td>good</td><td>fair</td><td>Bad</td></tr><tr><td>Probability (prior)</td><td>0.2</td><td>0.4</td><td>0.4</td></tr></table>	$p(x/w_i)$	good	fair	bad	Interesting lecture	0.8	0.5	0.1	Boring lecture	0.2	0.5	0.9	Quality of the course	good	fair	Bad	Probability (prior)	0.2	0.4	0.4	3									
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4.	Given the data points in the table beside, construct a decision tree using a purity threshold of 100%, i.e., the node is to be split if it contains different classes of examples and splitting is to be stopped when a leaf node is 100% pure with respect to having the same class of examples. Note that Age is a numeric attribute and Car is a categorical attribute. Risk gives the class label for each point: high (H) or low (L). Use information gain as the split point evaluation measure. Analyse each step clearly. <table><tr><td>x_i</td><td>Age</td><td>Car</td><td>Risk</td></tr><tr><td>x_1</td><td>25</td><td>Sports</td><td>L</td></tr><tr><td>x_2</td><td>20</td><td>Vintage</td><td>H</td></tr><tr><td>x_3</td><td>25</td><td>Sports</td><td>L</td></tr><tr><td>x_4</td><td>45</td><td>SUV</td><td>H</td></tr><tr><td>x_5</td><td>20</td><td>Sports</td><td>H</td></tr><tr><td>x_6</td><td>25</td><td>SUV</td><td>H</td></tr></table>	x_i	Age	Car	Risk	x_1	25	Sports	L	x_2	20	Vintage	H	x_3	25	Sports	L	x_4	45	SUV	H	x_5	20	Sports	H	x_6	25	SUV	H	3	CO 2,3 L5,6
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(B)	Random Forest is often preferred over a single decision tree. Analytically explain how feature randomness and bagging combine to reduce overfitting.	3																													
5.	The optimization problem that SVM tries to solve is $\min_{a,b} W^T W$ $s.t. \forall i y_i (W^T X_i + b) \geq 1$																														
(A)	Derive the maximum-margin classification problem reduces the above optimisation problem.	3	CO3 L4,5,6																												
(B)	How the SVM address the problem if the condition $\forall i y_i (W^T X_i + b) \geq 1$ does not hold	3																													